

Regrouping the merged expansion by exponent

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Abstract

The merged $d = 2$ cutoff expansion, indexed by face indices $i < M_1$, $j < M_2$, is regrouped over the finite pole set $\Lambda_M = \{\alpha_i\}_{i < M_1} \cup \{\delta_j\}_{j < M_2}$ as $\sum_{\alpha \in \Lambda_M} N^{-\alpha} (A_\alpha \log N + B_\alpha)$ with the canonical cutoff-independent coefficients of the previous unit (`mergedSum_eq_canon`). The compatibility inequalities $(M_1 - 1)/k_1 < M_2/k_2$ and $(M_2 - 1)/k_2 < M_1/k_1$ are used exactly once each: a retained pole of one axis lies strictly below every discarded pole of the other, so no canonical coefficient is supported outside the retained ranges. The analytic cutoff theorems follow in canonical pole form (`twoD_cutoff_canon`, `twoD_cutoff_amplitude_canon`). Zero sorry/axiom.

1 The things formalised

```
theorem uExp_lt_vExp_of (p : ) (h h k k M M : ) (hk : 0 < k) (hk : 0 < k)
  (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p)
  (hM : ((M : ) - 1) / k < (M : ) / k) (i j : ) (hi : i < M) (hj : M j) :
  uExp h k i < vExp h k j
theorem not_uPole_of_vExp ... (j : ) (hj : j < M)
  (hno : i Finset.range M, uExp h k i vExp h k j) :
  i, uExp h k i vExp h k j
theorem uface_gamma_lt_of ... (i : ) (hi : i < M) : (k : ) * uExp h k i - h - 1 < M
noncomputable def poleSet (h h k k M M : ) : Finset :=
  (Finset.range M).image (uExp h k) (Finset.range M).image (vExp h k)
theorem canonC_sum_eq ... :
  poleSet h h k k M M, transferTerm 1 (canonC h h k k c ) N
  = mergedLog h h k k M M c N
theorem mergedSum_eq_canon ... (hU : i, i < M, canonical FP = FP at order M)
  (hV : j, j < M, canonical FP = FP at order M) (N : ) :
  mergedSum b h h k k M M a b j c N
  =
  poleSet h h k k M M,
  N ^ (-) * (canonA h h k k c * Real.log N
  + canonB b h h k k a b j c )
theorem twoD_cutoff_amplitude_canon ( b p : ) (h h k k M M : ) (h : 0 < )
  (hb : 0 < b) (hb : b < ) (hk : 0 < k) (hk : 0 < k) (hp : ((h : ) + 1) / k = p)
  (hp : ((h : ) + 1) / k = p)
  (hM : ((M : ) - 1) / k < (M : ) / k) (hM : ((M : ) - 1) / k < (M : ) / k)
  (x y : * → ) (hx : WSummable x) (hy : WSummable y) :
  (fun N : => twoDAMP b N h h k k (anaAmp (ampCoeff x y) b)
  -
  poleSet h h k k M M,
  N ^ (-) * (canonA h h k k (fun i j s => ampCoeff x y (i, j) s)
  * Real.log N
  + canonB b h h k k (anaFaceU (ampCoeff x y) b)
```

```

      (anaFaceV (ampCoeff x y) b) (fun i j s => ampCoeff x y (i, j) s) ))
=0[atTop] fun N : =>
  N ^ (-(p + min ((M : ) / k) ((M : ) / k))) * (1 + Real.log N)

```

2 Mathematics

Each canonical pole term splits into the u -face, v -face and collision contributions (`canon_term_eq`). The u -face block over Λ_M reduces to the block over the u -poles because U_α vanishes at a v -pole δ_j ($j < M_2$) which is no retained u -pole: it is then no u -pole at all, since $\alpha_i \geq \alpha_{M_1} = p + M_1/k_1 > p + (M_2 - 1)/k_2 \geq \delta_j$ for $i \geq M_1$. Over the u -poles the sum pulls back along the injective map $i \mapsto \alpha_i$ and the canonical finite part is replaced by the order- M_2 one (admissible since $\gamma_i = k_2\alpha_i - h_2 - 1 < M_2$, again by compatibility). The collision block is the merged log family: for each $i < M_1$ either a unique $j_0 < M_2$ collides (then the indicator sum is that single term and $C_{\alpha_i} = c_{ij_0}/(k_1k_2)$) or none does, in which case α_i is no v -pole at all (compatibility once more) and both sides vanish.

3 Paper-facing meaning

With this the analytic $d = 2$ theorem reads exactly like `thm:TaylorTree` restricted to a finite pole set: $Z(N) = \sum_{\alpha \in \Lambda_M} N^{-\alpha} P_\alpha(\log N) + O(N^{-q_M}(1 + \log N))$ with P_α of degree ≤ 1 and coefficients defined once for all. The remaining step to the paper’s “for every T ” form is the choice $M_\ell = \lceil k_\ell R \rceil$, $R = \max(1, 2T - p)$, which satisfies both compatibility inequalities and $q_M \geq 2T$, plus discarding the retained poles $\geq 2T$ into the remainder.

4 Lean notes

- `Finset.sum_subset` composed as `((Finset.sum_subset Finset.subset_union_left ?_).symm).trans ?_` turns a sum over a union into a sum over one part plus a vanishing obligation; then `Finset.sum_image` with the axis injectivity pulls back to `range`.
- The `if` conditions in `mergedLog` are spelled with the literal exponent expressions; a `change` to the `uExp/vExp` form (definitional) is needed before `if_pos/if_neg` with hypotheses stated via `uExp`.
- `gcongr` proves $i/k_1 \leq (M_1 - 1)/k_1$ from the side goal $i \leq M_1 - 1$ (discharged by `linarith` from the cast of $i + 1 \leq M_1$).
- `lt_div_iff` is the current name for clearing a positive denominator in a strict inequality.
- The unit compiled on the first attempt.